

NGUYEN BA LONG

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Summary

Computer Science student interested in AI/ML, NLP, and RAG systems. I have built academic and personal projects using Python, FastAPI, PyTorch, Scikit-learn, LangChain, and LangGraph, including Vietnamese NLP applications, course video retrieval systems, and reinforcement learning experiments. I am looking for an AI/ML Engineer Intern role where I can apply my project experience, improve my engineering skills, and contribute to real-world machine learning systems.

Education

University of Information Technology – VNUHCM

Sep. 2023 – Feb. 2027 (Expected)

Bachelor of Science in Computer Science

Ho Chi Minh City, Vietnam

- **Relevant Coursework:** Machine Learning, Deep Learning, Data Mining, Computer Vision, Natural Language Processing, Software Engineering, Big Data.

Technical Skills

Languages: Python, Java, C/C++, SQL, JavaScript

Machine Learning & Deep Learning: PyTorch, TensorFlow/Keras, Scikit-learn, Hugging Face Transformers, BERT-based models, TF-IDF

NLP & RAG: Underthesea, LangChain, LangGraph, BM25, Vector Search, Hybrid Retrieval, Reranking

Backend & Databases: FastAPI, Flask, PostgreSQL, SQLAlchemy, ChromaDB, Elasticsearch

Tools: Docker, Docker Compose, Git, GitHub, Streamlit, Gradio, Jupyter Notebook

Projects

DashBot: A3C Dashboard Generation | *Python, PyTorch, FastAPI, A3C, JavaScript*

[GitHub](#)

- Reimplemented the DashBot paper for automatic dashboard generation from tabular CSV data using reinforcement learning.
- Built the core pipeline to analyze CSV files, detect useful insights, calculate rewards, and recommend dashboard charts through an A3C-based workflow.
- Implemented a PyTorch Bi-LSTM actor-critic model, A3C-style recommendation flow, greedy baseline, training scripts, pytest tests, and FastAPI endpoints.
- Developed a simple HTML/CSS/JavaScript frontend to upload CSV files and display recommended dashboard charts from the backend API.

Temporal RAG for UIT Course Video Retrieval | *FastAPI, LangGraph, ChromaDB, Elasticsearch*

[GitHub](#)

- Worked in a 3-member team to build a RAG-based Q&A system for learning materials, including text and video-related data.
- Implemented backend and retrieval workflows with FastAPI, PostgreSQL, LangChain, and LangGraph to support question answering over course materials.
- Combined ChromaDB/Elasticsearch vector storage, BM25 keyword search, and routing logic to retrieve useful learning content.
- Integrated Gradio UI, Docker Compose, Langfuse, and RAGAS to run, test, and evaluate the system more easily.

Vietnamese Hate Speech Detector | *Python, FastAPI, Scikit-learn, Underthesea*

[GitHub](#)

- Built an end-to-end Vietnamese NLP system to classify text into clean, offensive, and hate speech classes.
- Implemented text preprocessing, teencode expansion, punctuation removal, word segmentation with Underthesea, and TF-IDF-based features.
- Trained and evaluated machine learning models, reaching over 0.73 Macro F1 on the ViHSD dataset.
- Created FastAPI endpoints and a web interface for real-time text analysis and batch testing.

Self-Refined RL Reward Designer | *Python, Flask, Q-Learning, LLM API*

[GitHub](#)

- Reproduced a paper idea about using LLMs to generate and improve reward functions for reinforcement learning.
- Adapted the original robotics-based idea to a CPU-friendly 2D Gridworld environment with Q-Learning.
- Built Flask API and orchestration code for reward generation, agent training, evaluation, and self-refinement loops.
- Organized key modules for environment setup, Q-learning agent, LLM API caller, prompt files, and generated reward functions.

Languages

Vietnamese: Native

English: Basic working proficiency; TOEIC Speaking & Writing 250; currently improving English skills.